BCC BSIT Capstone Project Guide: Chapters 1-3 Documentation

Revised Academic Year 2024-2025

Chapter 1 - Introduction

- **Background of the Study**
  - Presentation of the importance of computing solutions—such as system software, applications, IT infrastructure plans, or other technologies—in enhancing operations and addressing challenges in organizations.
  - Research Locale
    ** Tell something about the organization (target beneficiary) where the research will be conducted. Content must be relevant to your study.*
    ** You may include historical information relevant to your study.*
    **This part must be in paragraph form also.*
  - Encountered Problems/Difficulties that will establish the need for the system.

    **State here the unsatisfactory condition/felt problem that needs a solution.*

    ** Ensure that the problems mentioned here are specific to the organization and can be effectively addressed by the software system to be developed. The content of this section must be based on the problems identified and presented during the title defense.*

    **Note that this part must be in paragraph form also.*
  - Introduction of the computing solution to be developed, presenting it as an improved approach to enhancing or replacing the organization's existing system.

- **Purpose and Description**

**State the purpose and description of the project in a way that it will highlight the significance of the project. This part must be written in general language and non-technical terms*

**This section must:*

1. *State the main purpose of the project*
2. *List all possible beneficiaries(Groups/Personalities/Organizations) and discuss how the system will benefit them.*
3. *State the rationale of the project.*

Sample:

Purpose and Description

This Capstone Project was conducted in order to _____

_____.

Once the proposed _____(title of the system)_____ is implemented to _____(target benefiary)_____, It will hold particular significance for the following beneficiaries:

1. *Beneficiary 1 – Explain how Beneficiary 1 will benefit after implementation of the computing solution.*
2. *Beneficiary 2 – Explain how Beneficiary 2 will benefit…*
3. *Beneficiary N – Explain how Beneficiary N will benefit…*

In the final part of this section, briefly explain the project's rationale based on its stated purpose and description. Then, include a statement assuming that the proposed computing solution will effectively address or resolve the existing problems or challenges.

- **Objectives of the Study**

  The main objective of the study is to design and develop a _____(*name of system*)_____for ____(*organization/company*)_____ . . . . .

  Furthermore, the developers aim to achieve the following specific objectives:

  1. To analyze the existing process of _____*(important process/processes that requires analysis for the development of the system*)_____ in the organization to identify inefficiencies, challenges, and opportunities for improvement.

     FOR EXAMPLE in an enrollment system the Objective 1 would be:

     - *To analyze the existing process of enrolment in the school to identify inefficiencies, challenges, and opportunities for improvement.*

     In a sales and inventory system, the Objective 1 would be:

     - *To analyze the existing sales transaction and inventory management process of the business to identify inefficiencies, challenges, and opportunities for improvement.*

  2. To identify the features of the system for the following users:
     - *User type 1*
     - *User type 2*
     - *User type 3*
     - *User type n*

     Example in an Enrollment System:

     To identify the key features of the system for different user roles, including:

     - *Students*
     - *Faculty/Teachers*
     - *Principal*

3. To test and evaluate the system's functionality, performance, security, usability and acceptability to ensure it meets user requirements and standards.
4. To prepare an implementation plan for the deployment of system.

- **Conceptual Framework**

***** Begin by providing a brief introduction to the Input-Process-Output (IPO) model, explaining its purpose. Clearly define how the inputs, process, and expected output contribute to the overall development of the project.

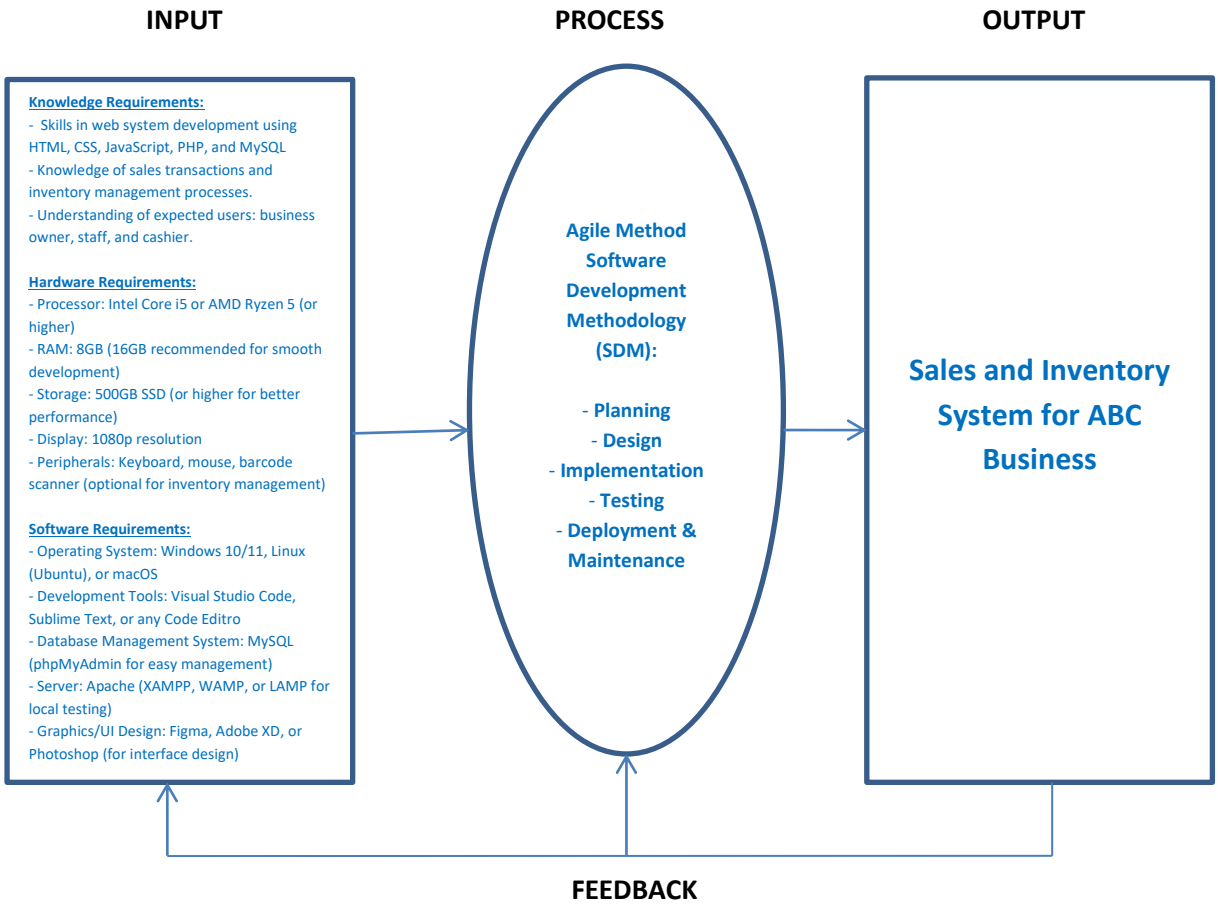
***** Create and include the Input-Process-Output illustration

- Input includes **knowledge requirements**(*technical skills and domain knowledge*), **hardware requirements (***components needed for both development and deployment***)** and **software** requirements (software tools such as development environments, database management systems, and frameworks).

-Process refers to the Software Development Methodology to be used (indicate its phases)

-Output is the expected computing solution as the result of the development process.

Sample Illustration:



* The IPO illustration in this guide is provided as an example. You may design your own layout as long as it includes the key elements of the IPO that accurately represent the conceptual framework.

* End this conceptual framework section with a brief discussion explaining the contents of the Input-Process-Output (IPO) illustration.

- **Scope and Limitations**

  * This section outlines the coverage and constraints of the project, as defined during the title defense. It specifies the software/system's functionalities and operational boundaries while also identifying aspects that are beyond its scope.

  Scope

  *Content must be in paragraph form.

  * This section must:

  1. Mention the platforms, frameworks, and tools utilized in development.
  2. Mention the key functionalities that the software/system will provide.

  Limitations

  *Content must be in paragraph form.

  * This section must:

  1. Mention functionalities that will not be included in the software/system.
  2. Define restrictions (if any) related to performance, security, or scalability
  3. Indicate any restrictions in terms of user roles, locations, or required devices.

- **Definition of terms**

  List all key/important terms in the study and clearly defined it based on how it was used in the study.

- **Review of Related Literature**

  Sources and Publication Date Requirements:

  - Local Studies – At least 5 or more sources published from 2020-2025.
  - Foreign Studies – At least 5 or more sources published from 2020-2025.

  Guidelines for Topic Selection:

  - The topics and parameters of the Review of Related Literature must be directly aligned with the Objectives of the Study to ensure relevance and coherence. Literature that does not contribute to addressing the study's objectives should not be included.

- Each related literature must support one or more specific objectives of the study. Avoid relying on only one aspect of the study; instead, cover multiple relevant parameters that collectively contribute to achieving the research goals.

For example, if a study focuses on developing a Sales and Inventory System, the objectives may include:

1. Identifying the existing process of sales and inventory management.
2. Determining the essential features for the system.
3. Conducting testing and evaluation to the system.
4. Preparing an implementation plan for system deployment

In this case, the related literature should include topics such as:

1. **Existing Sales and Inventory Management Processes** – A review of traditional and modern methods.
2. **Key Features of Effective Business Systems** – Studies on functionalities required by users.
3. **Usability and Acceptability of Information Systems** – Research on testing methodologies and user experience.
4. **System Implementation and Deployment Strategies** – Literature on best practices for rolling out new software systems in organizations.

You may include any review from a related study provided that the review:

- Directly support the study's Statement of Objectives.
- Provide background and historical developments related to the research focus.
- Identify gaps or areas for further investigation to justify the study's significance.
- Serve as evidence for the potential positive outcomes of the study.
- Analyze strengths and weaknesses of existing solutions to improve the study's direction.
- Guide the research process, ensuring alignment between literature, objectives, and methodology.

For best results, you must use any of the following search engines in searching for research materials online:

a. Google Scholar (https://scholar.google.com/)
b. Microsoft Academic (https://academic.microsoft.com/home)
c. BASE (https://www.base-search.net/)
d. CORE (https://core.ac.uk/)
e. CiteseerX (https://citeseerx.ist.psu.edu/index)
f. Science.gov (https://www.science.gov/)
g. Springer Open (https://www.springeropen.com/)

h. ScienceOpen.com(https://www.scienceopen.com/)
i. Sage Open(https://journals.sagepub.com/home/sgo

Note:

The literature review contains a number of "mini reviews" based on the materials read. Every mini review contains the following important parts:

- Brief summary of the related research material- this includes the brief description of the research presented in the literature related to your research subject. You can also include a brief introduction/statement that will establish the connection of your research subject/topic to the material being reviewed.

- Critically evaluated points in the research material related to your research subject for a particular parameter.

- Name of the author and year published or written (citation)
  Example:
  Willard (2018)
  Derwing et al., 2020

Chapter 2 – Methodology and Design

- Begin by including statements that briefly introduce the chapter and its contents.

- **Software Development Methodology**
  * Begin by explaining the significance of SDM in software engineering. Relate it also to your study.
  * Second, state the type of SDM used, and explain why the SDM was chosen.
  * Third, Clearly define the selected SDM, providing its key characteristics, principles, and typical use cases.
  *Fourth, Include a visual representation(illustration image) of the chosen SDM.
  * Fifth, Break down the phases of the SDM and explain the specific tasks that developers completed during each phase.

- **Sources of Data**
  ** These are **individuals**, **groups**, or **locations** that hold particular importance as **data** sources for researchers.*

- **Data Gathering Techniques**

  - This section should provide a clear explanation of how data will be collected.

  ** List and explain each technique used, including **HOW**, **WHEN**, and **WHY IT WAS APPLIED.***

  The common data gathering techniques are as follows:

  Surveys and Questionnaires

  Interviews

  Observation

  Document Analysis

  Library research

- **System Design**

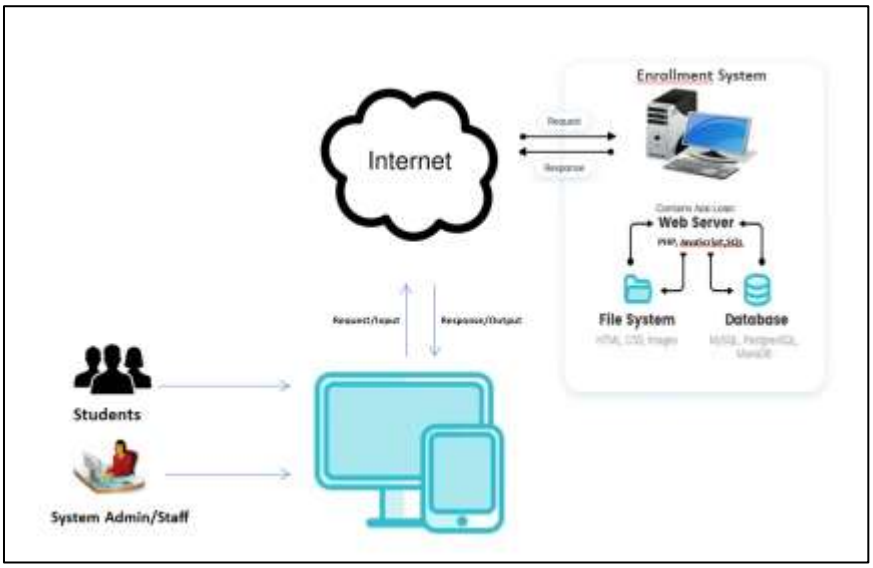
  - System Architecture (High-level system structure overview)

    - Begin by describing the diagram's importance and purpose

    - Present System Architecture Diagram.

    - Discuss the System Architecture Diagram.

    Sample System Architecture Illustration:



  - Existing Process Flowchart (If applicable, showing the manual/current process before automation)

    ***The existing process must be illustrated using flowchart.**

    - Begin by defining flowchart and describing its importance and purpose

    - Present Flowchart Diagram.

    - Discuss the content of flowchart.

- **Dataflow Diagram (DFD)**

  - Begin by defining DFD and describing its importance and purpose

  - Present DFD illustration.

  - Discuss the content of DFD.

  *** Please note of the following important components of DFD:**

  -External Entities (rectangle)

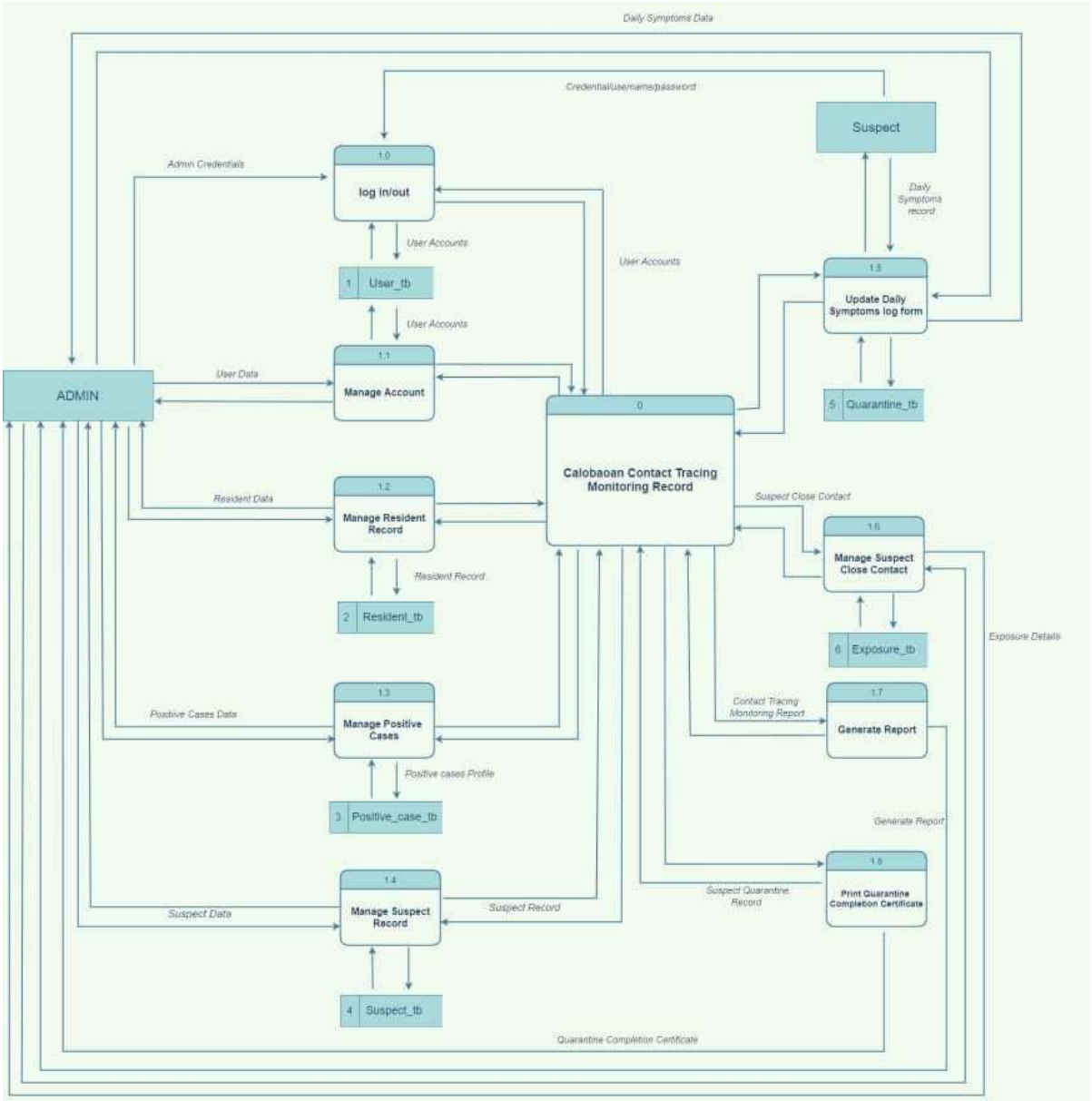
  - Process (square with rounded corners with process number at top)

  - Data Store

  - Data Flow Lines (in and out)

  - Main System (big square with rounded corners)

Sample DFD:



- **Entity-Relationship Diagram(ERD)**

  - Begin by defining ERD and describing its importance and purpose
  - Present ERD illustration.
  - Discuss the content of ERD.

  *** Please note of the following important components of ERD:**

  - **Entities** (uppercase, singular form)
  - **Attributes** - These are the characteristics of entities, represented within the entity box. Some attributes serve as **Primary Keys** (underlined and placed first in the attribute box) or **Foreign Keys**, which establish relationships between entities.

  - **Relationship** (typically 1 to many relationship, cardinality maybe mandatory or optional, with label)

- **Implementation Plan**

  * Implementation Plan must include the following contents:

  - **Project Timeline**

    - Presents the development schedule, key milestones, and expected completion dates for each project phase.

    - The timeline must be illustrated using a **Gantt Chart** for clear visualization.

  - **Deployment Plan** – Details the process for system installation, testing, and launch, including deployment strategies and user integration.
  - **Resource Requirements** – Specifies the necessary hardware, software, and human resources essential for successful implementation.

Chapter 3 – Results and Discussion

In the beginning of this chapter, briefly introduce what this chapter will cover.

- **Proposed System Flowchart**

  * Include a diagram of the system's workflow to illustrate how different components interact.
  * Briefly describe each step or process in the flowchart to ensure clarity.

- **System Features and User Interfaces**

  * List and explain the core features of the system.
  * Provide screenshots of the User Interface (UI) with a short description of each page/module.

- **System Testing and Evaluation**

  Functional Testing

  Purpose: To verify if the system's features work as intended based on requirements.
  How to Execute:

  1. **Create Test Cases** – Define multiple test scenarios that specify the expected behavior or outcomes for each system feature.
  2. **Perform Manual Testing** – Execute the test cases by interacting with the system as an end-user.
  3. **Document the Results** – Record whether each test case **Passed or Failed**, noting any errors or unexpected behaviors for further debugging and improvement.

  * Start by explaining the significance of functional testing, emphasizing its role in ensuring that the system's features operate correctly according to requirements.

  * List and describe the test cases used in the evaluation, specifying the expected outcomes for each scenario.

  * Describe the functional testing process.

  Performance Testing

  Purpose: To evaluate how the system performs under various conditions, such as speed, responsiveness, and load capacity.
  How to Execute: Create Test Cases, Perform Manual Testing and then Document the Results

* Start by explaining the significance of performance testing, emphasizing its role in ensuring that the system operates efficiently under different conditions, including varying workloads and user demands.

* List and describe the test cases used in the evaluation, specifying the expected outcomes for each scenario.

* Describe the performance testing process.

Security Testing

Purpose: To evaluate how the system performs under various conditions, such as speed, responsiveness, and load capacity.
How to Execute: Create Test Cases, Perform Manual Testing and then Document the Results

* Start by explaining the significance of security testing, emphasizing its role in ensuring that the system is protected against vulnerabilities, unauthorized access, and potential cyber threats.

* List and describe the test cases used in the evaluation, specifying the expected outcomes for each scenario.

* Describe the security testing process.

Usability Testing

Purpose: To verify the system's user-friendliness and ease of navigation.
How to Execute:

1. **Design a Usability Survey Form** – Create a structured questionnaire that assesses key usability aspects, such as ease of use, system efficiency, and interface design. Use a **Likert Scale (1-5 rating)** to measure user satisfaction.
2. **Conduct User Interaction** – Allow expected users (e.g., target audience or stakeholders) to explore and interact with the system. Ensure they perform essential tasks to simulate real-world usage.
3. **Collect User Feedback** – Ask participants to complete the survey form based on their experience. Encourage additional comments for qualitative insights.

* Start by explaining the significance of usability testing, emphasizing its role in ensuring that the system is user-friendly, intuitive, and accessible to its intended users.

* Explain the evaluation criteria, for example: ease of use, interface design, system responsiveness, and user satisfaction.

* Describe the usability testing process.

User Acceptance Testing (UAT)

Purpose: To verify if the system meets stakeholder expectations.
How to Execute: Design an Acceptability Survey Form, Conduct User Interaction and then Collect User Feedback

* Start by explaining the significance of acceptance testing, highlighting its role in ensuring that the system meets the specified requirements and expectations of end users and stakeholders before deployment.

* Define the key evaluation criteria, for example: system functionality, reliability, usability, performance, and compliance with user needs.

* Describe the acceptance testing process, including test planning, execution of real-world scenarios by target users, gathering feedback, and validating whether the system meets the required standards for final approval.

- **Implementation Results**

* Explain the deployment process, whether in a test environment, a live setting, or through pilot testing.

* Include issues encountered during implementation and how they were resolved.

- **Analysis of Results**

* Present the Analysis for Functional Testing

Sample Analysis Format for **Functional Testing**:

Test Case	Expected Output	Actual Output	Pass/Fail	Remarks
Login with correct credentials	User is redirected to the dashboard	Works as expected	Pass	N/A
Submit empty form	System shows validation error	No validation error	Fail	Add required field validation
Generate monthly sales report	PDF is generated successfully	Report crashes	Fail	Optimize report generation

* After presenting the analysis briefly describe the system's success rate in Functional Testing.

Calculate the success rate using the formula:

Success Rate = (Number of Passed Tests / Total Tests) × 100%

* Present the Analysis for Performance Testing

Sample Analysis Format for **Performance Testing**:

Test Scenario	Expected Time (seconds)	Actual Time (seconds)	Pass/Fail	Remarks
Load homepage (1 user)	< 3 sec	2.5 sec	Pass	N/A
Process a transaction while the database table contains 10,000 records	< 5 sec	10 sec	Failed	Optimize database queries
50 users accessing reports	< 8 sec	15 sec	Failed	Consider caching for faster reports

* After presenting the analysis, briefly describe the system's success rate in Performance Testing.

Calculate the success rate using the formula:

Success Rate = (Number of Passed Tests / Total Tests) × 100%

* Present the Analysis for Security Testing

Sample Analysis Format for **Security Testing**:

Security Test	Expected Behavior	Actual Behavior	Pass/Fail	Remarks
Login with wrong password	Show "Invalid credentials"	Works as expected	Pass	N/A
SQL Injection attempt in login	System rejects malicious input	Logs in without password	Fail	Use parameterized queries
View admin	Access denied	Can access	Fail	Fix role-based

dashboard as a normal user		admin panel		access

* After presenting the analysis, briefly describe the system's success rate in Security Testing.

Calculate the success rate using the formula:

Success Rate = (Number of Passed Tests / Total Tests) × 100%

* Present the Analysis for Usability Testing

Sample Analysis Format for **Usability Testing**:

Evaluation Criteria	**Strongly Disagree (1)**	**Disagree (2)**	**Neutral (3)**	**Agree (4)**	**Strongly Agree (5)**	**Average Score**
The system is easy to navigate	1	2	5	6	6	3.8
The interface design is clear and visually appealing	0	3	4	7	6	3.9
System instructions and labels are understandable	0	2	3	8	7	4.0
The system is user-friendly and requires minimal effort to learn	0	1	3	10	6	4.1

* After presenting the analysis, briefly describe the overall average usability rating of the system.

Calculate the **overall average usability rating** using the formula:

Rating = Total Average Score in all Criteria / Total Number of Criteria.

In the given example, the **overall average usability rating is 3.95**

* Present the Analysis for User Acceptance Testing

Sample Analysis Format for **UAT**:

Evaluation Criteria	**Strongly Disagree (1)**	**Disagree (2)**	**Neutral (3)**	**Agree (4)**	**Strongly Agree (5)**	**Average Score**
System functionality works as expected	0	2	3	8	7	**4.0**
The system is easy to navigate	1	2	5	6	6	**3.8**
System performance is fast and responsive	0	3	4	7	6	**3.9**
The system meets my needs and requirements	0	1	3	10	6	**4.1**

* After presenting the analysis, briefly describe the overall average acceptance rating of the system.

Calculate the **overall average acceptance rating** using the formula:

Rating = Total Average Score in all Criteria / Total Number of Criteria.

In the given example, the **overall average acceptance rating is 3.95**

- **Discussion of Findings**

* Include here the statements regarding the interpretation of results in Functionality, Performance, Security, Usability and Acceptance Testing.
* Include here the statements that highlight the system's strengths and areas for further development.

Important Notes for Chapter 3 Submission (Capstone 1: Chapters 1-3)

For the Capstone 1 Chapter 1–Chapter 3 Submission, follow these guidelines for Chapter 3 content:

Provide a brief overview of what this chapter 3 will cover.

Proposed System Flowchart

- Not required for this submission.

System Features and User Interface Discussion

- List and describe the system features, ensuring they are organized per user role (e.g., Admin, Staff, Customer).
- User Interface (UI) is not required yet—focus only on feature descriptions.

System Testing and Evaluation

- Define each type of testing (e.g., Functional, Performance, Security, Usability, and Acceptance Testing).
- Describe the testing plan for each type, outlining the intended approach but without actual testing results at this stage.

Implementation Results

- Not required for this submission.

Analysis of Results

- Not required for this submission.

Discussion of Findings

- Not required for this submission.